Alexander Wang

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Project Portfolio: https://wangalex8.github.io/
Lexington, MA (Permanent) | West Lafayette, IN (Current)

Education

- **Purdue University, College of Engineering** — West Lafayette, IN
  Bachelor of Science in Mechanical Engineering; GPA: 4.0 — *August 2025 - May 2029*

Experience

- **TRACE Laboratory** — West Lafayette, IN
  Undergraduate Research Assistant — *January 2026 - Present*
  - Fabricating a lightweight, 6-axis capacitive force-torque sensor for human-robot interaction and designing a custom mechanical device for sensor calibration in SolidWorks for ground-truth data validation.
  - Leveraging MuJoCo Playground to train a 29-DOF bipedal humanoid simulation model, optimizing for stable gait dynamics via proximal policy optimization (PPO) reinforcement learning frameworks.
  - Developing a technical foundation in machine learning for robotics through the implementation of predictive algorithms within a PyTorch framework.
- **Humanoid Robot Club Purdue** — West Lafayette, IN
  Mechanical Engineer — *September 2025 - Present*
  - Designed and fabricated a 2-degree-of-freedom humanoid wrist mechanism using SolidWorks, incorporating linear actuators and ball-joint linkages.
  - Collaborated on the modeling, design, fabrication, and testing of a humanoid upper-body system with an emphasis on mechanical stability and precision.
- **EPICS - Assistive Technology** — West Lafayette, IN
  Design Lead — *August 2025 - May 2026*
  - Led the mechanical development and optimization of an assistive mechanical horse that simulates the controlled movement of a real horse for individuals unable to participate in traditional hippotherapy.
  - Redesigned and fabricated robust wooden chassis components to replace non-durable parts, utilizing woodworking machinery (miter saws, drill presses) to improve durability and safety.

Personal Projects

- **Handheld Coaxial Contra-Rotating Fan**: Designing a compact fan system featuring coaxial contra-rotating blades driven by a single motor through a custom bevel gearbox. Used SolidWorks to model the fan blades, the fan housing, the bevel gears, the motor mounting bracket and the retaining plate. Created a SolidWorks assembly and implemented ball-bearings, custom spacers, and set-screws. Currently designing the fan's electrical system.
- **1-DOF Pure Rigid Body Robot Finger Mechanism**: Independently designed and modeled a novel 1 degree of freedom self-curling robotic finger mechanism comprised of 7 rigid body links. Fine-tuned linkage geometry and verified the mechanism's motion in SolidWorks.
- **Inverse Kinematics Solver using Supervised Machine Learning**: Developed a supervised learning inverse kinematics solver using PyTorch for a 2-DOF robot arm. Used OOP in python to build a forward kinematics simulator to calculate end-effector positions from 1,200 joint angles. Optimized a neural network utilizing nn.Linear layers, ReLU activation, and the Adam optimizer, achieving a 95.36% prediction accuracy on the test set after 1,200 training epochs.
- **MuJoCo Quadruped Walking Gait Simulation**: Simulated stable forward locomotion of a 12-DOF Quadruped Robot in a high-fidelity physics environment and implemented a time-varying PD controller with piecewise LERP trajectory mapping.

Skills

- **Software:** SolidWorks, Siemens NX, Autodesk Fusion, Siemens Teamcenter, Excel
- **Programming & Simulation:** Python, PyTorch, MATLAB, MuJoCo
- **Languages:** Chinese, English